Temporal Graph Models for Bankruptcy Prediction: A Fiscal-Year Evaluation

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Abstract. We propose Delta, a GCN–LSTM model that combines a firm's recent financial history with financial-peer information, and evaluate Charlie, a validation-selected fusion of LSTM and boosted-tree scores. On the 2019–2021 fiscal-year holdout, Charlie with XGBoost raises average precision (AP) from 0.114 to 0.137; its ROC-AUC changes from 0.887 to 0.890. The paired company-clustered interval excludes zero for AP but includes zero for ROC-AUC. Gains are not consistent on an earlier holdout. Exact filing dates are unavailable, so these results do not establish filing-time performance.

I. Introduction

Corporate bankruptcy is costly to firms, investors, employees, and lenders. Predictive models must find a small number of future events among many non-events, making average precision (AP) and recall at an operating point more informative than accuracy. Prior work on publicly traded U.S. firms reports that gradient-boosted trees are strong one-year-ahead predictors, establishing a demanding tabular baseline [1]. Graph neural networks (GNNs) have also been applied to bankruptcy and distress prediction, including models that combine corporate relations with temporal encoders [2], [3].

This study evaluates whether financial-peer graph messages improve bankruptcy ranking beyond firm-level temporal and boosted-tree models. Delta encodes up to five years of company features with an LSTM, then applies graph convolution over a financial-similarity peer graph. We also introduce Charlie, a validation-selected score fusion that tests whether the LSTM contributes information complementary to tuned XGBoost and LightGBM models. The comparison includes conventional graph models and a reliability-aware temporal GNN. Published results are reported separately when their datasets and evaluation protocols differ. The score fusion is an empirical combination method, not a new recurrent architecture.

II. Related Work

Altman introduced a discriminant score for bankruptcy prediction [4]. Alanis et al. benchmarked eight methods on U.S. public firms using walk-forward one-year-ahead forecasts and report ROC-AUC 0.92 for XGBoost [1]. Table III lists this result separately because the cohort, predictors, and protocol differ from those used here. Zhao et al. combine intra-risk and contagion-risk signals with graph learning [2]. Sun et al. combine relation-specific graph convolution, relation attention, temporal encoding, and score fusion for distress prediction [3]. These studies establish temporal and relational graph learning as prior approaches to financial distress prediction.

The American Stock Market bankruptcy release initially considered for this project has a target constant within every company across its observed years, so it cannot support the intended next-year forecasting task as released [5]. We instead use the public SEC-derived numeric extension and its inherited next-year bankruptcy labels [6]. The thesis that introduced this extension reports stronger AP for its FinBERT text model than for its numeric-only variants; those published figures are context, not results from our run, and are not directly comparable with our numeric-only experiment.

III. Data and Task

A. Source and Cohort

The source contains filing-level rows joined by company identifier and SEC accession, a binary label, a year, and 34 dense numeric financial concepts. Its reported full cohort is 42,956 filings from 2008–2021. We audited the downloaded dense variant and found 214 rows belonging to repeated company-year groups (103 groups); one repeated group had conflicting labels. Since the release does not provide filing dates, selecting an amendment or duplicate as the canonical filing would require an unsupported temporal assumption. We therefore exclude every row in a repeated company-year group. The resulting cohort contains 42,742 observations, 5,612 firms, 293 positive observations (0.6855%), and 34 numeric predictors. Source rows and license files are not redistributed by this project; readers should download them from the repository record and cite the dataset author [6].

The source defines its inherited target as bankruptcy in the next fiscal year. Exact filing or publication dates are unavailable, and 103 repeated company-year groups cannot be ordered by fiscal year alone. The evaluation therefore uses fiscal-year splits and does not verify whether each feature was available on a specific prediction date.

B. Chronological Split

We train on fiscal years 2008–2016, select models and fusion weights on 2017–2018, and evaluate once on 2019–2021. The test set contains 9,032 rows and 64 positive labels (0.7086% prevalence). Company identifiers occur in multiple years, so the primary split tests future years for a partially overlapping issuer population; the present report does not claim company-disjoint generalization. No test

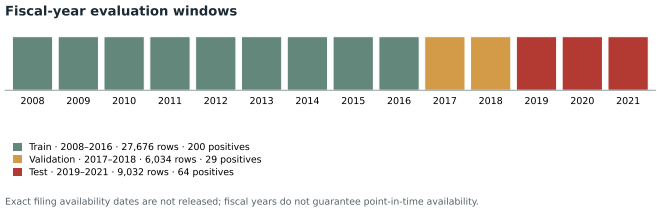


Fig. 1. Fiscal-year evaluation windows. The source does not provide exact filing-availability dates, so the split is chronological by fiscal year rather than verified filing time.

labels are used for feature transforms, model selection, or threshold selection.

IV. Methods

A. Preprocessing and Strong Tabular Baselines

Numeric features are converted to finite values, winsorized at the training-set 1st and 99th percentiles, median-imputed, and scaled by the training interquartile range. All transformation parameters are fitted on training years. We compare logistic regression, decision tree, random forest, histogram gradient boosting, MLP, LSTM, GCN, GraphSAGE, reliability-aware temporal GNN, LightGBM, and XGBoost. For each GBDT family we evaluate eight declared configurations on both the most recent-year input and flattened five-year history (16 validation trials), then select by validation AP. LightGBM selects 300 trees, learning rate 0.03, seven leaves, maximum depth four, 20 minimum child samples, and ℓ_2 penalty 1.0, with balanced class weighting. XGBoost selects depth two, learning rate 0.03, 300 trees, minimum child weight 10, ℓ_2 penalty 1.0, row and feature subsampling 0.8, and training-set positive-class weighting.

B. Delta: GCN with an LSTM Encoder

Delta encodes a trailing sequence of up to five annual feature vectors for firm i through an LSTM, producing state $\mathbf{h}_{i,t}$. At prediction year t, we construct a k-nearest-neighbor financial-similarity graph ($k = 10$) from the standardized financial features. The graph convolution combines each firm's temporal state with the similarity-weighted mean of its neighbors' states, followed by a binary classifier for next-year bankruptcy. Delta uses the conventional peer graph without relation-persistence weighting. Financial-similarity links represent predictive associations, not causal or contagion relationships.

C. Reliability-Aware Temporal GNN Comparator

For firm i in year t, an LSTM maps a trailing sequence of up to five annual feature vectors to state $\mathbf{h}_{i,t}$. For each financial feature group r, we construct a directed k-nearest-neighbor graph ($k = 10$) using cosine similarity of the corresponding standardized features in year t. For edge (j, i), the base weight is $s_{ji,t} = \max(0.01, 1 - d_{\cos}(\mathbf{x}_{j,t}, \mathbf{x}_{i,t}))$. We down-weight unstable neighborhoods using the Jaccard overlap $p_{i,t}$ between firm i's peer sets in adjacent snapshots:

$$w_{ji,t}^{(r)} = s_{ji,t}^{(r)} \left(0.5 + 0.5 p_{i,t}^{(r)}\right). \tag{1}$$

Each relation channel computes a weighted mean of neighboring temporal states. A learned softmax gate combines relation messages, which are fused with the firm's own state and passed to a binary classifier. The GCN and GraphSAGE comparators use the same LSTM states and contemporaneous peer graph, with a single merged message channel. Financial similarity edges are predictive associations; they do not establish contagion or causal links.

D. Graph Score Fusion Baselines

We transform the tree and graph scores with robust location/scale parameters fitted on validation data, then evaluate convex combinations $q = (1 - \alpha)q_{\text{tree}} + \alpha q_{\text{graph}}$ for $\alpha \in \{0, 0.1, \ldots, 1\}$. We choose the combination with highest validation AP subject to validation ROC-AUC not falling below the tree score. The same validation-only rule selects $\alpha = 0.2$ for LightGBM–GraphSAGE, $\alpha = 0.1$ for LightGBM–reliability-GNN, and $\alpha = 0.1$ for XGBoost–reliability-GNN. XGBoost–GCN and XGBoost–GraphSAGE select $\alpha = 0$, so validation does not support adding either message-passing score under this rule.

E. Charlie: LSTM–GBDT Score Fusion

Charlie combines the multi-head LSTM score with a boosted-tree score. For each model, we center scores at the validation median and divide by the validation 95th-to-5th percentile range. For the tree and LSTM scores z_{tree} and z_{LSTM}, Charlie computes

$$q_{\text{Charlie}} = (1 - \alpha)z_{\text{tree}} + \alpha z_{\text{LSTM}}. \tag{2}$$

We select α from $\{0, 0.1, \ldots, 1\}$ by validation AP, subject to validation ROC-AUC being no lower than the tree model. The late window uses the validation-selected four-year raw-feature multi-head LSTM with missingness channels; the early window uses a five-year multi-head attention LSTM. Both LSTM scores average five final seeds. We apply validation-fitted scaling and the selected weight to test scores without using test labels. Since this is score fusion, Charlie is evaluated as a ranking method and has no Brier score.

F. Metrics and Uncertainty

AP is the primary metric and ROC-AUC is co-required. We also report Brier score for individual probability models and precision/recall at a threshold chosen on validation to target 80% recall. Fused standardized scores are rankings, not calibrated probabilities; we do not report Brier scores for them. Paired 95% confidence intervals compare test predictions by resampling companies with replacement and retaining all test-year rows for each sampled company. Existing graph-fusion comparisons use

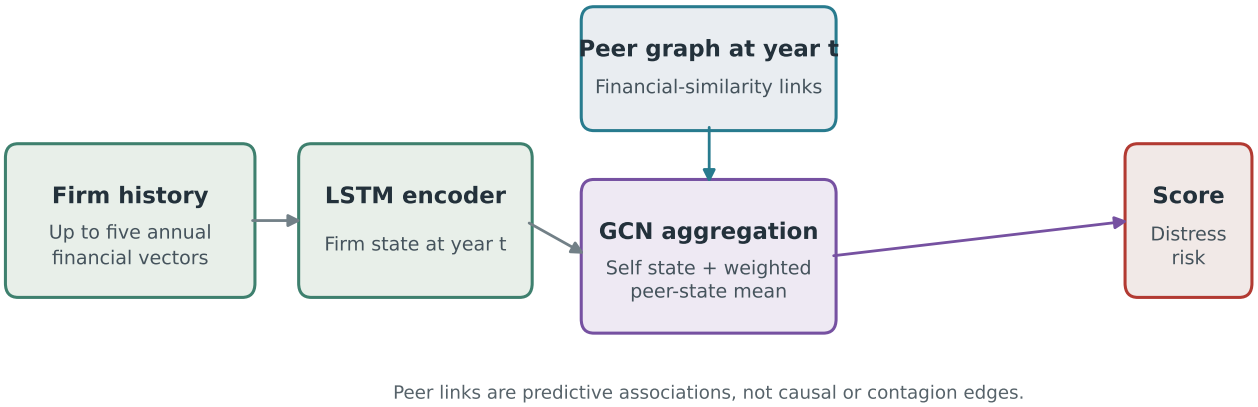


Fig. 2. Delta combines an LSTM encoding of up to five annual financial vectors with graph convolution over the financial-similarity peer graph at year t. The GCN combines a firm's temporal state with a similarity-weighted mean of peer states before predicting distress risk.

2,000 replicates; Charlie comparisons use 10,000. Intervals do not capture seed or period variability.

V. Experimental Results

Table III preserves the published XGBoost benchmark value. It is context only, not a matched test-set result.

XGBoost is the strongest individual baseline in this evaluation on both AP and ROC-AUC, with AP 0.1141 and ROC-AUC 0.8868. Delta achieves AP 0.0794 and ROC-AUC 0.7708. Its AP is 0.0086 higher than the LSTM's, while its ROC-AUC is 0.0032 lower. The reliability-aware temporal GNN comparator achieves AP 0.0383 and ROC-AUC 0.7515. The graph-model comparisons use one training seed; Charlie uses five-seed mean LSTM scores. Figure 3 visualizes every metric in the main comparison, while Figure 4 shows ranking curves for the tabular baselines and temporal graph models.

The validation-selected LightGBM–GraphSAGE blend increases test AP from 0.1116 to 0.1181, while ROC-AUC decreases from 0.8790 to 0.8705. The paired company-clustered AP difference is 0.00515 with 95% interval [–0.00285, 0.01913]; the ROC-AUC difference is –0.00869 [–0.01783, –0.00065]. For XGBoost, validation selects a GNN weight of 0.1. The fused test AP is 0.1133 and ROC-AUC is 0.8806, compared with 0.1141 and 0.8868 for XGBoost alone. The paired AP-difference interval is [–0.01363, 0.01493], and the ROC-AUC difference is –0.00626 [–0.01558, –0.00037].

Charlie with XGBoost increases AP from 0.1141 to 0.1375 on the 2019–2021 holdout, with a paired mean difference of 0.0233 and 95% interval [0.0021, 0.0518]. ROC-AUC changes from 0.8868 to 0.8895; its paired interval, [–0.0021, 0.0078], includes zero. Charlie with LightGBM reaches AP 0.1300 and ROC-AUC 0.8875, but the paired intervals include zero for both metrics. On the earlier 2015–2016 holdout, AP increases for both tree models by point estimate, but both AP intervals include zero. XGBoost–Charlie ROC-AUC decreases from 0.8601 to 0.8497, and all earlier-window ROC-AUC intervals include zero. Figure 5 and Table II report paired changes and both fiscal-year windows. These repeated observations are exploratory because the same dataset and model-development process informed both windows.

An additional diagnostic adds 12 conventional financial ratios to the 34 source features. Ratio-augmented LightGBM scores AP 0.0827 and ROC-AUC 0.8670. The ratio-augmented reliability-GNN scores AP 0.0286.

The originating dataset thesis reports AP 0.08 and ROC-AUC 0.87 for its raw numeric XBRL representation, and AP 0.30 and ROC-AUC 0.82 for its text FinBERT system [6]. These results use the unreduced filing cohort and are included as source-reported context.

VI. Conclusion

Delta combines an LSTM representation of firm history with graph convolution over financial-similarity peers. Charlie tests whether a validation-selected LSTM score adds signal to boosted-tree rankings. The late-window XGBoost fusion improves AP by point estimate and its paired interval excludes zero, while its ROC-AUC interval includes zero. Earlier-window gains are uncertain and the XGBoost fusion has lower ROC-AUC by point estimate. The experiments therefore do not establish consistent improvement on both required metrics. Exact filing dates were unavailable, so the comparisons use fiscal-year ordering.

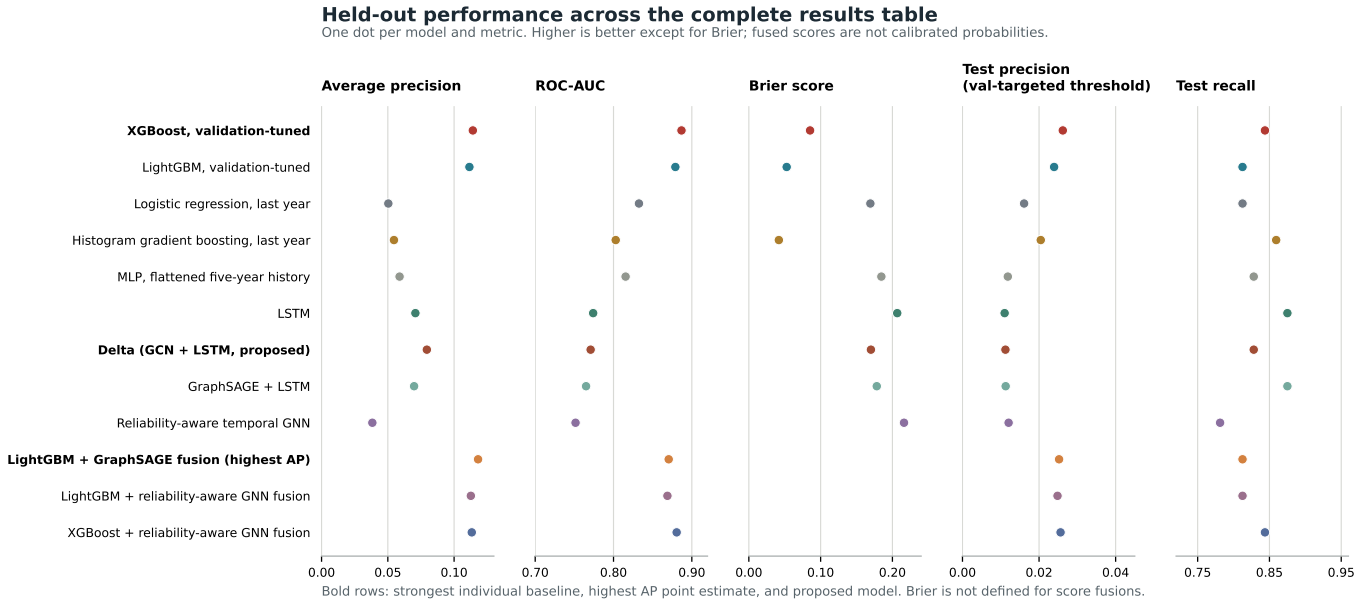


Fig. 3. All metrics in Table I, shown as metric-specific panels on their native scales. Higher is better except for Brier score. Score-fusion rows have no Brier score because the fused ranking scores are not calibrated probabilities. The strongest individual baseline and proposed model are bolded.

TABLE I

Held-out fiscal-year test results (2019–2021, $n = 9{,}032$, 64 positives). Thresholds target 80% recall on validation; precision and recall are measured on test. Bold marks the strongest individual baseline, highest AP point estimate, and proposed model.

Model	AP ↑	ROC-AUC ↑	Brier ↓	Test precision	Test recall
XGBoost (best individual baseline)	0.1141	0.8868	0.0853	0.0262	0.8438
LightGBM, validation-tuned	0.1116	0.8790	0.0527	0.0239	0.8125
Logistic regression, last year	0.0504	0.8325	0.1693	0.0161	0.8125
Histogram gradient boosting, last year	0.0546	0.8028	0.0418	0.0204	0.8594
MLP, flattened five-year history	0.0589	0.8154	0.1846	0.0118	0.8281
LSTM	0.0708	0.7740	0.2067	0.0110	0.8750
(Ours) Delta (GCN + LSTM)	0.0794	0.7708	0.1701	0.0112	0.8281
GraphSAGE + LSTM	0.0698	0.7649	0.1782	0.0113	0.8750
Reliability-aware temporal GNN	0.0383	0.7515	0.2162	0.0120	0.7812
LightGBM + GraphSAGE fusion (highest AP)	0.1181	0.8705	–	0.0252	0.8125
LightGBM + reliability-aware GNN fusion	0.1127	0.8688	–	0.0248	0.8125
XGBoost + reliability-aware GNN fusion	0.1133	0.8806	–	0.0256	0.8438

TABLE II

Charlie score-fusion results on two fiscal-year holdouts. The LSTM weight α is selected on validation only. Differences and 95% intervals compare paired test predictions with company-clustered bootstrap resampling (10,000 replicates).

Window	Tree model	α	Tree AP / ROC-AUC	Charlie AP / ROC-AUC	ΔAP [95% CI]
2015–2016	XGBoost	0.2	0.0826 / 0.8601	0.1036 / 0.8497	0.0234 [–0.0038, 0.0578]
2015–2016	LightGBM	0.8	0.0697 / 0.8536	0.0928 / 0.8580	0.0254 [–0.0552, 0.0974]
2019–2021	XGBoost	0.2	0.1141 / 0.8868	0.1375 / 0.8895	0.0233 [0.0021, 0.0518]
2019–2021	LightGBM	0.3	0.1116 / 0.8790	0.1300 / 0.8875	0.0175 [–0.0006, 0.0410]

Paired ΔROC-AUC [95% CI] for the rows above are, respectively, −0.0102 [−0.0460, 0.0193], 0.0045 [−0.0405, 0.0475], 0.0027 [−0.0021, 0.0078], and 0.0085 [−0.0004, 0.0193]. The late-window XGBoost AP interval excludes zero; none of the ROC-AUC intervals exclude zero.

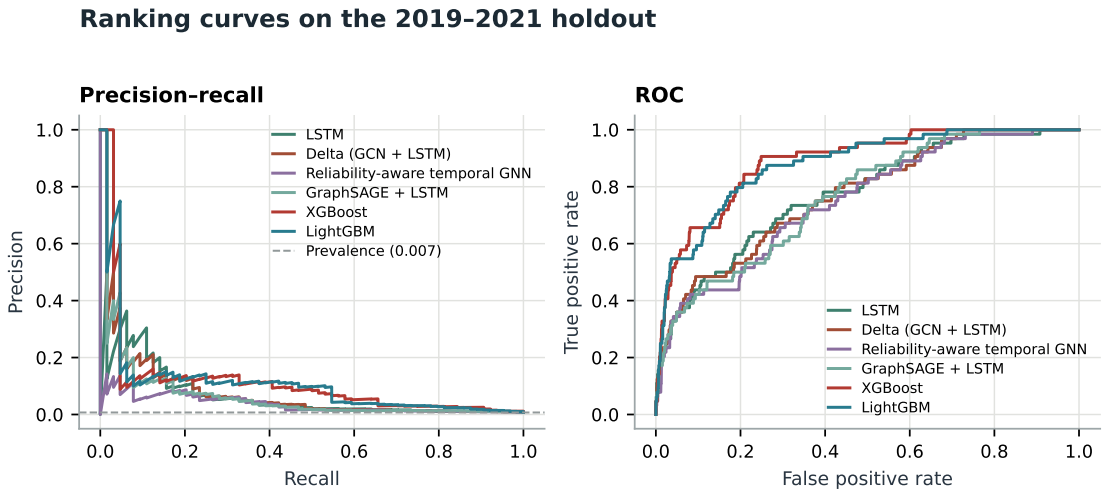


Fig. 4. Precision–recall and ROC curves for the strongest tabular baselines, LSTM, Delta (GCN–LSTM), GraphSAGE–LSTM, and the reliability-aware temporal GNN comparator on the held-out years. The dashed line in the precision–recall panel marks test prevalence. Curves use saved test-set predictions.

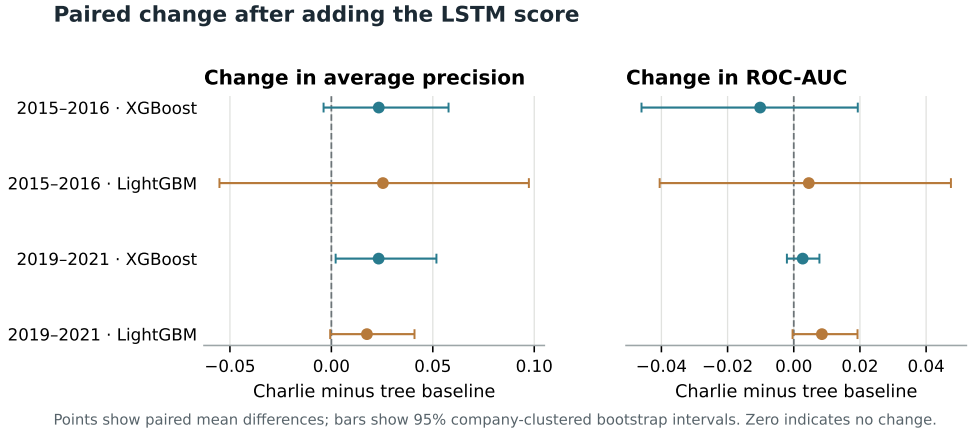


Fig. 5. Paired company-clustered changes in AP and ROC-AUC after adding the LSTM score to XGBoost or LightGBM. Points show mean differences and bars show 95% bootstrap intervals.

TABLE III
Published benchmark context from Alanis et al. The source reports ROC-AUC for XGBoost; AP is not reported. Different cohort, predictors, and walk-forward protocol make this non-comparable to our holdout.

Evaluation	AP	ROC-AUC
Alanis et al. [1]	n/r	0.9200
This study, XGBoost	0.1141	0.8868

References

[1] E. Alanis, S. Chava, and A. Shah, "Benchmarking machine learning models to predict corporate bankruptcy," arXiv preprint arXiv:2212.12051, 2022. [Online]. Available: https://arxiv.org/abs/2212.12051

[2] Y. Zhao, S. Wei, Y. Guo, Q. Yang, X. Chen, Q. Li, F. Zhuang, J. Liu, and G. Kou, "Combining intra-risk and contagion risk for enterprise bankruptcy prediction using graph neural networks," Information Sciences, 2023.

[3] Z. Sun, F. Shao, and F. Meng, "Network-aware financial distress prediction via hierarchical attention on dynamic heterogeneous corporate networks," Procedia Computer Science, vol. 287, pp. 453–460, 2026.

[4] E. I. Altman, "Financial ratios, discriminant analysis and the prediction of corporate bankruptcy," The Journal of Finance, vol. 23, no. 4, pp. 589–609, 1968.

[5] G. Lombardo, M. Pellegrino, G. Adosoglou, S. Cagnoni, P. M. Pardalos, and A. Poggi, "Machine learning for bankruptcy prediction in the american stock market: Dataset and benchmarks," Future Internet, vol. 14, no. 8, p. 244, 2022.

[6] A. Potiriadis, "An open financial dataset for bankruptcy prediction," Master's thesis, University of Peloponnese, Jul. 2026. [Online]. Available: https://doi.org/10.26263/amitos-3539